

READ ME for PINN work.

Two solutions to run it on the cluster:

- either on della, because A100 GPUs are available but it take roughly several hours to wait for the queue
- or use neuronic.cs (ryne also needs to sponsor you for that), and L40 GPUs which are also working (maybe a bit less than A100s), and there is usually no time to wait

For myself, I usually run script by using a python environment(torch-env) including PyTorch.
For example :

```
(torch-env) [gm7719@neuronic pinn-fiona]$ cd ~/pinn-fiona && sbatch --gres=gpu:1
--mem=64G --time=6:00:00 --job-name=obs_seq --output=obs_seq_%j.log --wrap="source
~/bashrc && conda activate torch-env && python -u pinn_fiona_3D_Tq_seq_obs.py
--use_obs 1 --mu 0.15 --nu 0.35 --n_tdr 5000 --n_ds 500 --n_data 5000 --n_colloc 10000
--gamma 0.5 --sigma_ffc 0 --lambda1 0 --n1_adam 60000 --n1_lbfgs 0 --lambda2 0.2
--n2_adam 20000 --n2_lbfgs 0 --lambda3 0 --n3_adam 20000 --n3_lbfgs 50000
--refresh_every 499 --stp 500 --M 100 --seed 0 --resampling_data_adam_1 1000
--resampling_eq_adam_1 1000 --resampling_obs_adam_1 1000
--resampling_data_adam_2 1000 --resampling_eq_adam_2 1000
--resampling_obs_adam_2 1000 --resampling_eq_adam_3 1000 --resampling_obs_adam_3
1000"
```

It usually takes between 1h and 2h depending on the number of epoch.

Even though it's not the best solution, everything is on the same .py.

The data used is a file where the pressure interpolation is already done [put the correct name].